

Ex. No. 3	Advanced SQL	
Date:		

Aim

To execute the given commands making use of aggregate functions, group by clause and order by clause.

Description:

Aggregate Functions:

They take a collection of values and return a single value as result. The various aggregate functions are:-

1)Sum: - takes a collection of values and returns the sum of the values.

Eg: select sum(sal) from emp;

2)Avg:- returns the average of the values.

Eg: select avg(sal) from emp;

3)Count:- returns the number of elements in the collection.

Eg: select count(*) from emp;

4)Min:- returns the minimum value in a collection.

Eg: select min(sal) from emp;

5)Max:- returns the maximum value in a collection.

Eg: select max(sal) from emp;

The input to sum and average must be a collection of numbers, but the other operators can operate on collections of non-numeric data types, such as strings as well. The average function will return the average of the given tuple. The aggregation function count is used frequently to count the number of tuples in relation. The for this function in SQL is count (*).

Distinct Keyword

To eliminate the duplicates the keyword distinct is used in the aggregation expression. Sql does not allow the use of distinct with count(*). It is legal to use distinct with max and min, even

though the result does not change.

Eg: Select count (distinct job);

GROUP BY Clause: -

To apply aggregate function to a group of sets of tuples. The attributes given in the group by clause are used to form groups. Tuples with some value on all attributes in the group by clause are placed in one group.

Eg: Select branch_name, avg(bal) from account group by branch_name;

ORDER BY Clause: -

This clause causes the tuples in the result of a query to appear in sorted order. We specify asc for ascending order and desc for descending order.

Eg: Select * from loan order by amount desc;

Sample Input/Output:

1. List the max ,min and average salary in the company.

SQL> select sum(salary) as total_salary , min(salary) as min_salary, max(salary) as max_salary , avg (salary) as avg_salary from Emp;

TOTAL_SALARY	MIN_SALARY	MAX_SALARY	AVG_SALARY
164000	7000	50000	23428.5714

2. List the max salary paid to a salesman.

SQL> select max(Salary) as max_salary from Emp where Job_Name='Salesman';

MAX_SALARY
10000

3. List the total salary , max ,min and average salary of employee according dept no.20. SQL> select sum(salary) as total_salary , min(salary) as min_salary, max(salary) as max_salary , avg (salary) as avg_salary from Emp where dept_no=20;

TOTAL_SALARY	MIN_SALARY	MAX_SALARY	AVG_SALARY
60000	10000	50000	30000

4. List the no of employee and average salary for employee in Dept no.20.

SQL> select count(*) , avg(salary) as avg_salary from Emp where dept_no=20;

COUNT(*) AVG_SALARY

230000

QUESTIONS:

1. List the number of users in the user table.
2. List the number of events conducted during 2023.
3. Calculate the total price for the confirmed tickets.
4. Find the minimum, maximum and average time taken for the given events.
5. Retrieve distinct number of users.
6. List names of events was more than 10 hours.
7. List the seat numbers of the users those whose tickets are confirmed.
8. List user name and phone numbers those whose tickets reserved.
9. List the user details in the ascending order of their ticket price.
10. Modify the ticket price of the user based on the following:
 - a. If the price is greater than 30, give the concision of 2%.
 - b. If the price is lesser or equal to 30, give the concision of 4%.
11. List the Venue name and address in each country.
12. List all the city name in Alphabetical order.

RESULT: